

Vaishnavi Kouru - software engineer

+1 650 305 5350 | vkouru07@gmail.com | [portfolio](#) | linkedin.com/in/vaishnavi-kouru | [github](#) | U.S. Citizen

SKILLS

Languages & Frameworks: Rust, C/C++, Python3, Java, TypeScript, SQL, Bash, RESTful APIs, Firebase, Flask, SocketIO, Numpy, React, Node

Tools & Concepts: Ubuntu Linux, Unix, MacOS

EDUCATION

B.S. Computer Science, San Jose State University

2023 – 2027

Courses: Formal Languages and Computability (CS 154), Programming Paradigms (CS 152), Concepts of Compiler Design (CS 153), Individual Studies for Honors (CS 180H), Data Structures and Algorithms (CS 146), OOD (CS 151), Computer Systems (CS 47), Operating Systems (CS 149), Computer Architecture (CS 147), Information Security (CS 166), Big Data Processing (CS 131)

Activities: Member of Society of Women Engineers, Competitive Programming Club

Positions: Grader for CS 154, Computer Science Study Lab Tutor

EXPERIENCE

(Modalix Edge AI Development Kit) Industry Project

Apr 2026 – Present

SiMa.ai — San Jose

- Designed and implemented "Little ACT," a hardware-constrained Action Chunking Transformer policy (64×64 inputs, TinyCNN encoder, 2+2 transformer layers) restricted to a Modalix-compatible op set (Conv/Linear/MatMul/Softmax/LayerNorm/ReLU) with static shapes for edge deployment
- Built an imitation-learning data pipeline in NVIDIA Isaac Lab, generating cube-stacking demonstration trajectories (RGB images, proprioceptive joint states, action chunks) for policy training
- Trained the policy and exported to static-shape ONNX (opset 17); currently integrating SiMa Palette Model SDK for INT8 quantization and compilation to the Modalix target
- Brought up the DevKit from scratch (UART serial bring-up, Linux driver conflict resolution) and documented the full pipeline for reproducibility

Software Developer

April 2025 – Present

Midtown Family Services — San Jose

- Developed a tablet- and mobile-friendly Salesforce registration app for a community food pantry, capturing client contact info, household demographics, and USDA TEFAP affirmations.
- Created dynamic reports and dashboards tracking household visits, age demographics, and service program sources.
- Prototyped and tested features in a Salesforce Sandbox to protect production data.

ML Data Annotator

Dec 2023 – Present

DataAnnotation — Remote

- Trained AI/ML models regarding the accuracy of their code and language use to optimize model performance.
- Tools used: Pandas, Tabulate, Numpy, Supabase, Node JS, Langchain.

UX Extern

Jun 2022 – Jul 2022

Cisco — Remote

ID Tech Roblox Studio / Lua Coding Instructor

May 2025 – Aug 2025

ID Tech — Stanford University, Stanford, CA

Minecraft Python Coding Camp Instructor

May 2024 – Aug 2024

Brains and Motion — Sunnyvale, CA

PROJECTS

The End of the Track

React, Node, Flask, SocketIO, Firebase

- Built a web application hosting the game "The End of the Track" using React, Node + Flask, SocketIO, and Firebase.
- Handled Websocket connections between server and clients, assigned users to game rooms, listened for and emitted events.
- Features: synchronous multiplayer (live player vs. player), supports user login/signup or guest authentication.
- Designed the database schemas for user and game data, and original Figma design and artwork/sprites for the site.

Syllabi-Interpreter

JavaFX, StanfordNLP, ConvertAPI

- Built a desktop app that extracts and analyzes relevant information from syllabi and organizes it into actionable formats.
- Used StanfordNLP for interpreting unstructured data, and ConvertAPI for the PDF parsing.
- Designed formats for the data visualization and used JavaFX to implement them.
- Collaborated with 3 other teammates to create designs, models and implementation.